Sirius
The brightest nighttime star of all is Sirius (below center), in the constellation Canis Major. Its magnitude is –1.44. At 8.6 light-years away, it is also one of the closest stars to us. It gives out about 20 times more light than the Sun.

A piece of sky

The celestial sphere is divided into areas called constellations, each of which contains a figure produced by joining the brightest stars in dot-to-dot fashion. These figures are purely products of the human imagination—the stars have no real relationship to one another in space. The stars, as well as other objects within a constellation, are identified by name or number according to an agreed system of rules.

BRIGHTNESS

It is easy to observe that the stars in the sky vary in brightness. The first star-watchers soon realized this, and so they classed the stars according to their brightness level. Their system was later formalized to produce what is known as the apparent-magnitude scale used today.

Apparent magnitude is an indication of how bright a star appears when viewed from Earth. This is not the same as the star's real brightness, its luminosity. Each star is given a number, known as its magnitude, to indicate its brightness: the smaller the number, the brighter the star. The brightest stars have negative magnitude values.

The apparent-magnitude scale is also used for other objects— the full Moon is magnitude –12.5, for example. The brightness of a planet varies depending on its distance from the Sun and Earth; at its brightest, Venus measures –4.7.

TEN BRIGHTEST NIGHT-SKY STARS

The stars listed below are in order of brightness. Only the four brightest have negative values, and as the magnitude number increases, the star becomes fainter. All stars of magnitude 6 and brighter are visible with the naked eye, so each of these stars is easily seen.

1. **Sirius**
   Canis Major, –1.44
2. **Canopus**
   Carina, –0.62
3. **Rigil Kentaurus**
   Centaurus, –0.28
4. **Arcturus**
   Boötes, –0.05
5. **Vega**
   Lyra, 0.03
6. **Capella**
   Auriga, 0.08
7. **Rigel**
   Orion, 0.18
8. **Procyon**
   Canis Minor, 0.40
9. **Achernar**
   Eridanus, 0.45
10. **Betelgeuse**
    Orion, 0.45

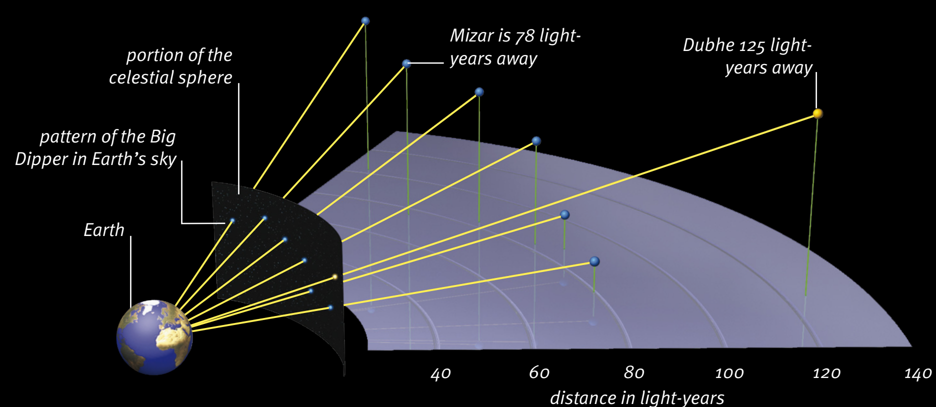
RELATIVE DISTANCE

The stars on the celestial sphere all appear to be at the same distance from Earth, but the reality is quite different. The stars are scattered through space with vast expanses between them. Each star is typically 7 light-years away from its nearest neighbor. And they are all so far from us that they appear to be pinpoints of light rather than huge globes.

The distance of the naked-eye stars we see, for instance, varies hugely. As we look at them, we see stars that are just a few light-years from us and others that are thousands of light-years away. This means that the star patterns of the constellations are illusions. They are a two-dimensional view of stars within a three-dimensional volume of space. The stars are not related at all, but far apart. Very often they are farther from each other than they are from Earth.

OWL NEBULA (M97), A PLANETARY NEBULA

BODE'S GALAXY (M81), A SPIRAL GALAXY



Line-of-sight effect
The stars in the constellation Ursa Major trace out the shape of a bear in Earth's sky. The stars that form the tail and rump of the bear are known as the Big Dipper. The stars are at vastly different distances from Earth, and seen from elsewhere in space, they would make a totally different pattern.

The Big Dipper
Also known as the Plow, the Big Dipper (left of center) consists of seven bright stars. Three stars form the handle, and four more the basin of a saucepan in profile. Once found, the shape can be used as a guide to other objects. The two brightest stars are known as the pointers, because they point the way to the Pole Star, Polaris.